Aditya Ghosh   Curriculum Vitae

Last updated: July 2026

Contact details

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Education

Ph.D. in Statistics, **Stanford University** GPA 4.13/4.00 — 2022 – 2027 (expected)

- Advised by Stefan Wager (Stanford GSB) & Dominik Rothenhäusler (Stanford Statistics)

Masters of Statistics (M.Stat), **Indian Statistical Institute**, Kolkata — 2020 – 2022

- **Dissertation advisor:** Prof. Bodhisattva Sen (Columbia University)

Bachelor of Statistics (B.Stat), **Indian Statistical Institute**, Kolkata — 2017 – 2020

Research

My research focuses on problems in Causal Inference. Currently working on Causal Inference in Dynamic Settings (with Stefan Wager) and Causal Inference under Distribution Shift (with Dominik Rothenhäusler).

5. **Ghosh, A.** & Wager, S. (2025+). Non-parametric Causal Inference in Dynamic Thresholding Designs. *Submitted.* arXiv *Preliminary version accepted at EC 2026*

   Summary: How to study the long-term effects of threshold-based interventions when units dynamically evolve over time? We show that simple reduced-form characterizations remain available for a dynamic marginal policy effect at the treatment threshold.

4. **Ghosh, A.** & Rothenhäusler, D. (2025+). Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies. *Submitted.* arXiv

   Summary: How to draw valid causal inference in observational studies when multiple covariate adjustments seem equally plausible? We provide valid inference for a reweighted population when at least one of the candidate adjustments is valid.

3. **Ghosh, A.**, Imbens, G. & Wager, S. (2025+). PLRD: Partially Linear Regression Discontinuity Inference. *Submitted.* arXiv R package

   Summary: For policies with an eligibility threshold, confidence intervals built using the existing methods often face a trade-off between nominal coverage and interval width. PLRD resolves this with bias-aware confidence intervals that are both valid and short.

2. **Ghosh, A.**, Deb, N., Karmakar, B., & Sen, B. (2026). Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A Design-based Perspective. *'Accepted' at Biometrika.* arXiv Journal link

   Summary: How to draw design-based causal inference using rank-based methods from nonparametric statistics? We study efficiency and robustness of Rosenbaum's rank-based estimator under finite population inference in randomized trials.

1. **Ghosh, A.** (2019). An asymptotic formula for the Chebyshev theta function. *Notes on Number Theory and Discrete Mathematics*, 25(4), 1-7. Journal link

   Summary: Derived sharp bounds on the geometric mean of the first n prime numbers.

Academic services

- I help organize the **Online Causal Inference Seminar**, a weekly seminar on causal inference with participants from all over the world.
- Served as a reviewer for: *AoS*, *JASA*, *JRSSB*, *NeurIPS*

Teaching

As instructor, **Stanford University** ExploreCourses

- Stats 302: Qualifying Exam Workshop (Theoretical Statistics). Summer 2025, 2026
- Stats 302: Qualifying Exam Workshop (Probability). Summer 2024

As teaching assistant, **Stanford University** ExploreCourses

- Stats 361: Causal Inference. Winter 2025
- Stats 310B/Math 230B: Theory of Probability II. Winter 2024
- Stats 310A/Math 230A: Theory of Probability I. Autumn 2023
- Stats 216: Introduction to Statistical Learning. Winter 2023
- Stats 209: Introduction to Causal Inference. Fall 2025
- Stats 202: Data Mining and Analysis. Summer 2023, Autumn 2022
- Stats 200: Introduction to Theoretical Statistics. Winter 2026, Autumn 2024
- Stats 141: Introduction to Statistics for Biology. Spring 2026
- Stats 60: Introduction to Statistical Methods: Precalculus. Spring 2025

Other experiences

- Trained numerous high school students for mathematics olympiads and other math competitions; materials are available at ghoshadi.wordpress.com .

Accolades

- **Departmental Teaching Assistant Award**, Stanford University 2026
- **IMS 2025 International Conference on Statistics and Data Science (ICSDS)**, student travel award 2025
- **IISA conference 2025**, best poster award 2025
- **Industrial Affiliates Annual Conference at Stanford**, travel award 2024
- Selected for **Machine Learning in Economics Summer Institute** at University of Chicago Booth school of business 2024
- **ISIAA – J. K. Ghosh Memorial Gold Medal** (for outstanding performance in Masters of Statistics) 2023
- **ISIAA – Mrs. M. R. Iyer Memorial Gold Medal** (for best overall performance in Bachelor of Statistics) 2021
- **Nikhilesh Bhattacharyya Memorial Gold Medal** (for best performance in major (Statistics) in Bachelor of Statistics) 2021
- **Madhava Mathematics Competition**, selected among the top few participants in the country for a prestigious summer camp 2019, 2018
- **Indian National Mathematical Olympiad**, earned a prestigious certificate of merit from NBHM, Govt. of India (awarded to the top 75 participants in the country) 2016

Invited Talks

- **Joint Statistical Meeting**, Boston, Massachusetts 2026

  Title: Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies.
- **ACM Conference on Economics and Computation (EC'26)**, Rome, Italy 2026

  Title: Nonparametric Causal Inference in Dynamic Thresholding Designs
- **Metrics-lunch** (an weekly seminar on Econometrics), Stanford University 2026

  Title: Non-parametric Causal Inference in Dynamic Thresholding Designs

- **Data-driven Seminar**, Stanford University 2026

  Title: Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A Design-based Perspective
- **University of Washington** Causal Inference & Missing Data Reading Group 2026

  Title: Assumption-robust Causal Inference
- **IMS International Conference on Statistics and Data Science**, Seville, Spain 2025

  Title: Assumption-robust Causal Inference
- **Stanford Causal Science Center (SC^2) Conference**, Stanford University 2025

  Title: Causal Inference in Dynamic Thresholding Designs
- **Joint Statistical Meeting**, Nashville, Tennessee 2025

  Title: Causal Inference in Dynamic Thresholding Designs
- **Industrial Affiliates Annual Conference**, Stanford University 2024

  Title: Practical Bias-aware Inference in Regression Discontinuity Designs
- **Stanford Causal Science Center (SC^2) Conference**, Stanford University 2024

  Title: Asymptotic bias-aware inference in regression discontinuity designs under higher-order smoothness
- **Computational and Methodological Statistics**, Berlin, Germany 2023

  Title: Efficiency and robustness of Rosenbaum's regression (un)-adjusted rank-based estimator in randomized experiments
- **PCM Memorial Lecture**, Indian Statistical Institute, Kolkata 2022

  Title: The synthetic control method in causal inference
- **D. Basu Memorial Lecture**, Indian Statistical Institute, Kolkata 2021

  Title: Large low-rank matrix completion
- **Online Reading Group on Functional Data Analysis** 2021

  Title: Two-sample testing of the equality of mean functions
- **Students' Learning Seminar**, Indian Statistical Institute, Kolkata 2021

  Title: Matching estimators in causal inference

Class Projects

- **SMARTer Multi-task Fine-tuning of BERT**

  Summary: Engineered a multi-task NLP framework, integrating adversarial regularization, proximal optimization, and low-rank adaptation with randomized task scheduling for stable cross-task training. Joint with Disha Ghandwani and Rahul Kanekar.
- **Inference for Data Adaptively Sampled via REINFORCE**

  Summary: Developed inference methods for adaptively collected data and analyzed the robustness–efficiency trade-offs in variance estimation for long-horizon policy evaluation. Joint with Ivy Zhang.
- **Predicting the Number of Micro-businesses Across the U.S.**

  Summary: Built a forecasting pipeline using GoDaddy panel and Census data, applying ARIMA, GLMs, and varying-coefficient models to link demographics with county-level microbusiness dynamics. Joint with Joshua Kazdan and Yash Nair.

Skills

Python, **R**, Markdown, Git, C (basic), HTML (basic). **R package:** plrd